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Lead assembly for connecting solar panel arrays to inverter

Abstract

A lead assembly includes first and second feeder cables and multiple drop lines. The feeder cables each has a first or second diameter and a first or second polarity. Each of the feeder cables contains two or more joints. Each joint electrically couples one or more drop lines to the first or second feeder cable at a nexus. Each drop line has a diameter that is less than the first diameter and less than the second diameter. Each drop line coupled to the first feeder cable is configured to be electrically connected to a wire harness configured to electrically connect to an array of solar panels. Two or more of the droplines that are connected to the first feeder cable each includes an in line fuse disposed between the joint and the wire harness. Each joint has a mold enclosure configured to enclose at least a portion of the nexus.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 18/739,127, filed Jun. 10, 2024, which is a continuation of U.S. application Ser. No. 18/341,655, filed Jun. 26, 2023, which is a continuation of U.S. application Ser. No. 17/301,609, filed Apr. 8, 2021, which is a continuation of U.S. application Ser. No. 14/849,458, filed Sep. 9, 2015, which claims the benefit of and priority to U.S. Provisional application No. 62/047,773, filed Sep. 9, 2014. Each of the foregoing applications is incorporated herein by reference in its entirety.

BACKGROUND

(1) Disclosed embodiments relate to electrical components, and more specifically to assemblies for connecting solar panel arrays to an inverter, without the need of a combiner box.

(2) As depicted in FIG. 1, a conventional solar power configuration includes a plurality of solar panels **30** each connected to a central combiner box **36**, and a plurality of combiner boxes (not shown) connected to an inverter. The inverter converts the direct current (DC) generated by the solar panels into alternating power (AC), which goes in the “grid”.

(3) The configuration of FIG. 1 has been simplified in recent years by the wire harness of U.S. Pat. No. 8,604,342 which issued Dec. 10, 2013; and U.S. application Ser. No. 14/057,089 filed Oct. 18, 2013; and Ser. No. 14/296,726 filed Jun. 5, 2014, all of which are hereby incorporated by reference. More specifically, as shown in FIG. 2, wire harnesses **34** connect multiple solar panels **30** in an array **32**, so each array has a single input via wire harness **34**

into the combiner box. This has drastically reduced the number of connections going into a combiner box, which is safer, more reliable, easier to install, maintain and troubleshoot, and saves money over the conventional configuration of FIG. 1.

(4) Unfortunately, however, a combiner box is still a necessary component in known solar power installations because the energy coming from solar arrays must be combined prior to going into the inverter. Combiner boxes are problematic because they are clumsy, prone to damage and malfunctioning, must be periodically maintained, and require extensive planning and skill for installation. Due to the many connections going into a combiner box, a combiner box cannot easily be moved without disconnecting and reconnecting all the connections and wiring, which requires considerable manpower, danger, and expense.

(5) As can be seen, there is a need for a device that renders a combiner box unnecessary. It is desirable that this device is small, inexpensive to manufacture and transport, and easy to use. It is also desirable that the device is extremely durable and can be adapted for a variety of circumstances.

SUMMARY OF DISCLOSED EMBODIMENTS

(6) In an example embodiment, a lead assembly for use in a solar energy installation includes first and second feeder cables and multiple drop lines. The first feeder cable has a first diameter and a first polarity. The second feeder cable has a second diameter and a second polarity. Each of the first feeder cable and the second feeder cable contains two or more joints. Each joint electrically couples one or more drop lines to the first or second feeder cable at a nexus. Each drop line has a diameter that is less than the first diameter and less than the second diameter. Each drop line coupled to the first feeder cable is configured to be electrically connected to a wire harness configured to electrically connect to an array of solar panels. Two or more of the drop lines that are connected to the first feeder cable each includes an in line fuse disposed between the joint and the wire harness. Each joint has a mold enclosure configured to enclose at least a portion of the nexus.

(7) In another example embodiment, a solar power system includes a lead assembly, a first wire harness, first solar panels, a second wire harness, and second solar panels. The lead assembly includes first and second feeder cables and multiple drop lines. The first feeder cable has a first diameter and a first polarity. The second feeder cable has a second diameter and a second polarity. Each of the first feeder cable and the second feeder cable contains two or more joints. Each joint electrically couples one or more drop lines to the first or second feeder cable at a nexus. Each drop line has a diameter that is less than the first diameter and less than the second diameter. Two or more of the drop lines that are connected to the first feeder cable each includes an in line fuse disposed between the joint and the wire harness. Each joint has a mold enclosure configured to enclose at least a portion of the nexus. The first wire harness includes a first central trunk and first branches. The first solar panels are electrically coupled by the first wire harness to form a first solar array. Each branch of the first branches electrically couples a different one of the first solar panels to the first central trunk. The second wire harness includes a second central trunk and second branches. The second solar panels are electrically coupled by the second wire harness to form a second solar array. Each branch of the second branches electrically couples a different one of the second solar panels to the second central trunk. A first drop line of the drop lines terminates at a first drop line connector. The first drop line connector connects to the first central trunk. The first drop line connector and the first drop line electrically couple the first solar array to the first or second feeder cable. A second drop line of the drop lines terminates at a second drop line connector. The second drop line connector connects to the second central trunk. The second drop line connector and the second drop line electrically couple the second solar array to the first or second feeder cable.

(8) In another example embodiment, a lead assembly for use in a solar energy installation includes first and second drop lines, a feeder cable, an inline fuse, and a mold structure. The first drop line has a first diameter and terminates at a drop line connector. The drop line connector is configured to connect with a wire harness. The wire harness is configured to electrically connect to an array of solar panels. The second drop line has a second diameter. The feeder cable is electrically coupled to the first and second drop lines at a nexus. The feeder cable has a third diameter that is greater than the first diameter and greater than the second diameter. The inline fuse is disposed in the first drop line between the nexus and the drop line connector. The mold structure at least partially surrounds the nexus. The mold structure defines first, second, third, and fourth openings. The feeder cable extends from the first and second openings of the mold structure in opposite directions. A feeder cable passage extends completely through the mold structure between the first and second openings. The feeder cable extends through the feeder cable passage. The first drop line extends from the third opening of the mold structure. The second drop line extends from the fourth opening of the mold structure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 depicts a conventional wiring configuration with each solar panel individually wired to a central combiner box;

(2) FIG. 2 depicts an improved but known wiring configuration with solar panels harnessed together to form solar arrays, with each solar array individually wired to a central combiner box;

(3) FIG. 3 depicts the present invention of a plurality of wire harnesses coupled to a lead assembly that connects to an inverter, with the unnecessary combiner box shown for demonstration only;

(4) FIG. 4 schematically represents a system of the present invention;

(5) FIG. 5 depicts a positive lead assembly having a single drop line per joint;

(6) FIG. 6 depicts a negative lead assembly having a single drop line per joint;

(7) FIG. 7 schematically depicts a joint with a compression lug;

(8) FIG. 8 depicts a single drop joint;

(9) FIG. 9 depicts a configuration where a lead assembly has a single drop line as may be appropriate in use with crystalline panels;

(10) FIG. 10 depicts a configuration where a lead assembly has a single drop line as may be appropriate in use with thin film panels;

(11) FIGS. 11-12 depict the overmold for a dual drop lead assembly for 250 mcm cable;

(12) FIGS. 13-14 depict the overmold for a dual drop lead assembly for 750MCM cable;

(13) FIGS. 15-16 depict the undermold for a dual drop lead assembly for 250MCM cable;

(14) FIGS. 17-18 depict the undermold for a dual drop lead assembly for 750MCM cable;

(15) FIGS. 19-20 depict the overmold for a single drop lead assembly for 250MCM cable;

(16) FIGS. 21-22 depict the overmold for a single drop lead assembly for 750MCM cable;

(17) FIGS. 23-24 depict the undermold for a single drop lead assembly for 250MCM cable;

(18) FIGS. 25-26 depict the undermold for a single drop lead assembly for 750MCM cable;

(19) FIG. 27 depicts two lead assemblies with single drop, with the lower assembly including in line fuses;

(20) FIG. 28 depicts a spooled lead assembly with single drop;

(21) FIG. 29 depicts a dual drop lead assembly for 750MCM cable;

(22) FIG. 30 depicts a single drop lead assembly for 750MCM cable;

(23) FIG. 31 depicts a dual drop lead assembly for 750MCM cable with overmold shown;

(24) FIG. 32 depicts the overmold for a dual drop lead assembly for 750MCM cable;

(25) FIG. 33 depicts a dual drop lead assembly for 750MCM cable with undermold shown;

(26) FIG. 34 depicts the undermold for a dual drop lead assembly for 750MCM cable;

(27) FIG. 35 depicts a dual drop lead assembly for 750MCM cable with compression lug shown;

(28) FIG. 36 depicts the compression lug for a dual drop lead assembly for 750MCM cable;

(29) FIG. 37 depicts a dual drop lead assembly for 750MCM cable with wires exposed at nexus;

(30) FIG. 38 depicts the end piece for a dual drop lead assembly for 750MCM cable;

(31) FIG. 39 depicts a single drop lead assembly for 750MCM cable with overmold shown;

(32) FIG. 40 depicts the overmold for a dual single lead assembly for 750MCM cable;

(33) FIG. 41 depicts a single drop lead assembly for 750MCM cable with undermold shown;

(34) FIG. 42 depicts the undermold for a single drop lead assembly for 750MCM cable;

(35) FIG. 43 depicts a single drop lead assembly for 750MCM cable with compression lug shown;

(36) FIG. 44 depicts the compression lug for a single drop lead assembly for 750MCM cable;

(37) FIG. 45 depicts a single drop lead assembly for 750MCM cable with wires exposed at nexus;

(38) FIG. 46 depicts the end piece for a single drop lead assembly for 750MCM cable;

(39) FIG. 47 depicts a dual drop lead assembly for 250MCM cable;

(40) FIG. 48 depicts a single drop lead assembly for 250MCM cable with the overmold depicted in phantom;

(41) FIG. 49 depicts a dual drop lead assembly for 250MCM cable with overmold shown;

(42) FIG. 50 depicts the overmold for a dual drop lead assembly for 250MCM cable;

(43) FIG. 51 depicts a dual drop lead assembly for 250MCM cable with undermold shown;

(44) FIG. 52 depicts the undermold for a dual drop lead assembly for 250MCM cable;

(45) FIG. 53 depicts a dual drop lead assembly for 250MCM cable with wires exposed at nexus;

(46) FIG. 54 depicts the end of string for a dual drop lead assembly for 250MCM cable with the overmold depicted in phantom;

(47) FIG. 55 depicts a single drop lead assembly for 250MCM cable with the overmold depicted in phantom;

(48) FIG. 56 depicts the overmold for a single drop lead assembly for 250MCM cable;

(49) FIG. 57 depicts a single drop lead assembly for 250MCM cable with undermold shown;

(50) FIG. 58 depicts the undermold for a single drop lead assembly for 250MCM cable;

(51) FIG. 59 depicts a single drop lead assembly for 250MCM cable with wires exposed at nexus;

(52) FIG. 60 depicts the end of string for a single drop lead assembly for 250MCM cable with the overmold depicted in phantom; and

(53) FIG. 61 depicts a compression lug.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

(54) The following detailed description describes exemplary embodiments of the invention. The description is not to be taken in a limiting sense but is made merely for the purpose of illustrating the general principles of the invention, since the scope of the invention is best defined by the appended claims.

(55) The following structure numbers shall apply to the following structures among the various FIGS.: **10**—Lead assembly; **12**—Drop line; **13**—Drop line connector; **14**—Feeder cable; **15**—Feeder cable connector; **16**—Insulation; **17**—Joint; **18**—Exposed wire; **19**—Nexus; **20**—Compression lug; **22**—Undermold; **24**—Overmold; **25**—Aperture; **27**—Trunk buss jumper; **29**—In line fuse; **30**—Solar panel; **32**—Solar array; **34**—Wire harness; **35**—Wire harness connector; **36**—Combiner box; and **38**—Inverter.

(56) Referring to FIG. 3, a system of the present invention generally includes a plurality of solar panels **30** that are electrically coupled to form solar arrays **32a**, **32b**, and so forth. Panels **30** are preferably coupled using wire harnesses **34** which are commercially available from Shoals Technologies Group of Portland, TN. Each wire harness includes a plurality of branches each connecting to one solar panel, and a central trunk that terminates in wire harness connector **35**. Each wire harness is connected at wire harness connector **35** to drop line connector **13** (not shown) of lead assembly **10**.

(57) As shown in FIG. 3, solar arrays **32a**, **32b** and **32c** are connected to a lead assembly **10**. A single lead assembly **10** can reasonably accommodate approximately 200 Kw.

(58) While FIG. 5 depicts a positive lead assembly having in line fuses **29**, and FIG. 6 depicts a negative lead assembly without in line fuses, it should be understood that positive lead assemblies may lack in line fuses and negative lead assemblies may include in line fuses.

(59) Referring now to FIG. 5, lead assembly **10** generally includes drop line **12** joined to feeder cable **14** at joint **17**. As shown in FIG. 35, the drop line and feeder cable are held together by compression lug **20**, which is surrounded by undermold **22** (FIG. 33), which is surrounded by overmold **24** (FIG. 29). Overmold **24** preferably defines at least one aperture **25** for receiving zip-ties, and the like, for securing the assembly upon installation.

(60) Each drop line **12** terminates in drop line connector **13**, which connects drop line **12** to wire harness connector **35**. Drop line **12** is preferably constructed of 18 to 4 gauge wire, and drop line connectors **13** are preferably off-the-shelf connectors such as MC4/PV-KBT4/61-UR & PV-KST4/61-UR from Multi-Contact of Windsor, CA. Joint **17** of lead assembly **10** may include a single drop line **12**, as shown in FIG. 30, or dual drop lines **12**, as shown in FIG. 29, corresponding with connecting to a single wire harness or two wire harnesses, respectively. Whether a single or dual drop line is most suitable depends on the configuration of the solar arrays in a field.

(61) By way of example and referring to FIG. 10, in a solar installation having thin film panels, which occupy a large area of space, it may be advantageous to employ single drop assemblies because it is impractical to physically reach wire harnesses that are very spread out. Alternatively and referring to FIG. 9, in a solar installation having crystalline panels, which occupy a smaller area of space, it may be advantageous to employ dual drop assemblies because the associated wire harnesses are relatively close to one another and can be connected to a single lead assembly.

(62) It is also important to understand that the configuration of a particular lead assembly can be modified to accommodate different solar installations. For example, joints **17** and corresponding drop lines **12** can be spaced close together (approximately 15 cm), or far apart (approximately 15000 cm), along feeder cable **14**, depending on the density of solar panels. Also, spacing of joints **17** and corresponding drop lines **12** can vary on a single lead assembly.

(63) Each feeder cable **14** terminates in feeder cable connector **15**, which connects to trunk buss jumper **27** (FIG. 27), to inverter **38** (FIG. 3), or to connection unit, recombiner, or disconnect unit (not shown). Trunk buss jumper **27** is essentially an “extension cord” which is economical to use in some configurations, for example where portions of feeder cable **14** are installed at different times. In another situation, trunk buss jumper **27** could be buried underground and feeder cable **14** above ground. Being able to install these components independent of one another can offer much more diversity. Trunk buss jumper **27** could also be utilized if there are a significant number of varying lengths from solar array **32** to inverter **38**. Trunk buss jumper **27** could also be utilized if there is a substantial distance (greater than >50 meters) to travel from closest solar array **32** to inverter **38** and it would be wasteful to use a lead assembly with unused drop lines. The other end of feeder cable **14** terminates in joint **17** having end piece **26**, which is typically installed at the solar array located furthest from the inverter. In alternative embodiments, feeder cable **14** includes feeder cable connector **15** at both terminal ends so feeder cables **14** can be connected one-to-another in an end-to-end orientation. In yet another embodiment, one or both ends of feeder cables **14** are blunt cut for subsequent manual connection, for example stripping and crimping to connectors or other segments of feeder cable.

(64) Feeder cable **14** is preferably constructed of 6 gauge to 1000 MCM wire, with the specific wire chosen based on factors such as the number of associated drop lines and the distance between the connection and downstream inverter and whether or not feeder cable **14** is of aluminum or copper construction. Feeder cable connectors **15** are

preferably off-the-shelf connectors such as KBT10BV & KST10BV from Multi-Contact of Windsor, CA.

(65) Referring to FIG. 37, drop line 12 and feeder cable 14 electrically couple at nexus 19 of joint 17. This is preferably accomplished by stripping wire insulation 16 from corresponding segments of drop line 12 and feeder cable 14, adjoining respective segments of exposed wire 18, and securing contact between the segments of exposed wire by employing compression lug 20. It should be understood that securing contact between the segments could be achieved by other means including soldering, splicing, crimping, and so forth. Compression lug 20 is preferably surrounded by undermold 22, which is preferably composed of RTP 2099E×127663 from RTP Co. of Winona, MN that has been applied by injection molding. As shown in FIG. 48, undermold 22 is preferably surrounded by overmold 24, which is preferably composed of RTP 199×124807 from RTP Co of Winona, MN that has been applied by injection molding. The resulting assembly is profoundly durable, resistant to environmental factors such as temperature fluctuations, debris, and moisture, and is strong enough to be buried.

(66) In an alternative embodiment, there is no undermold, just an overmold, which is applied in a single molding process.

(67) It is a very important feature that a system of the present invention doesn't require a combiner box. Rather, lead assemblies 10 effectively “combine” the power from solar arrays 32 and deliver it to inverter 38. As shown in FIG. 3, the present invention also eliminates the need of DC feeders, which would carry power from the combiner box to the inverter in a conventional system. Also, the need for fuses can be eliminated by using drop lines including in line fuses 29, as shown in the lower assembly of FIG. 27. Appropriate fuses include of HP10x from Mersen of Newburyport, MA.

(68) An embodiment of the present invention preferably has the following specifications: Voltage rating of 600 VDC/1000 VDC/1500 VDC; Maximum branch current of 30 amps per string; Maximum overcurrent protection of 30 amps per string; maximum trunk cable size of 750MCM; and Maximum ambient operating temperature of 50° C., although other embodiments beyond these specifications are within the scope of the inventions.

(69) It should also be understood that the illustrations are for the purpose of describing a preferred embodiment of the inventions and are not intended to limit the inventions thereto. By way of example, the present invention can be scaled up or down to accommodate any feeder cable size, including the common 250 and 750MCM cable sizes. Also, while the current invention has been described in association with DC applications, it should be understood that it could also be implemented into feeders deployed in AC systems, which would negate the need for AC recombiner boxes. It should also be understood that approximations allow variances of +/-10%, unless otherwise noted or nonsensical such as a length less than zero. It should also be understood that all ranges set forth inherently include the endpoints themselves, as well as all increments, there between.

Claims

1. A lead assembly for use in a solar energy installation, the lead assembly comprising: a first feeder cable having a first diameter and a first polarity; a second feeder cable having a second diameter and a second polarity; and a plurality of drop lines; wherein: each of the first feeder cable and the second feeder cable contains two or more joints; each joint electrically couples one or more drop lines of the plurality of drop lines to the first or second feeder cable at a nexus; each drop line has a diameter that is less than the first diameter and less than the second diameter; each drop line coupled to the first feeder cable is configured to be electrically connected to a wire harness configured to electrically connect to an array of solar panels; two or more of the plurality of drop lines that are connected to the first feeder cable each includes an in line fuse disposed between the joint and the wire harness; and each joint has a mold enclosure configured to enclose at least a portion of the nexus; wherein a first mold enclosure that encloses at least a portion of a first nexus of the first feeder cable includes: a feeder cable passage that extends completely through the first mold enclosure, the first feeder cable extending through the feeder cable passage along a first longitudinal axis; a first drop line passage that extends at least partially through the first mold enclosure, a first drop line of the plurality of drop lines extending through the first drop line passage along a second longitudinal axis; and a second drop line passage that extends at least partially through the first mold enclosure, a second drop line of the plurality of drop lines extending through the second drop line passage along a third longitudinal axis; the first and second drop lines are part of a single drop line cable and the first and second drop line passages form a drop line cable passage that extends completely through the first mold enclosure.
2. The lead assembly of claim 1, wherein the first feeder cable is configured for electrical connection to an inverter without a combiner box between the first feeder cable and the inverter.
3. The lead assembly of claim 1, wherein each mold enclosure completely encloses the corresponding nexus.
4. The lead assembly of claim 1, wherein each drop line coupled to the first feeder cable is configured to be electrically connected to a central trunk of the wire harness, the central trunk having a plurality of branches that are configured to electrically connect to a separate solar panel contained in the array of solar panels.
5. The lead assembly of claim 1, wherein each mold enclosure comprises: an overmold that encloses the

corresponding nexus; or an undermold that encloses the corresponding nexus and an overmold that encloses the undermold.

6. The lead assembly of claim 1, wherein at least one of the second longitudinal axis or the third longitudinal axis is not parallel to the first longitudinal axis.

7. A solar power system, comprising: a lead assembly, the lead assembly comprising: a first feeder cable having a first diameter and a first polarity; a second feeder cable having a second diameter and a second polarity; and a plurality of drop lines; wherein: each of the first feeder cable and the second feeder cable contains two or more joints; each joint electrically couples one or more drop lines of the plurality of drop lines to the first or second feeder cable at a nexus; each drop line has a diameter that is less than the first diameter and less than the second diameter; two or more of the plurality of drop lines that are connected to the first feeder cable each includes an in line fuse disposed between the joint and a wire harness from a plurality of wire harnesses; and each joint has a mold enclosure configured to enclose at least a portion of the nexus; a first wire harness of the plurality of wire harnesses comprising a first central trunk and a first plurality of branches; a first plurality of solar panels electrically coupled by the first wire harness to form a first solar array, each branch of the first plurality of branches electrically coupling a different one of the first plurality of solar panels to the first central trunk; a second wire harness of the plurality of wire harnesses comprising a second central trunk and a second plurality of branches; a second plurality of solar panels electrically coupled by the second wire harness to form a second solar array, each branch of the second plurality of branches electrically coupling a different one of the second plurality of solar panels to the second central trunk; wherein: a first drop line of the plurality of drop lines terminates at a first drop line connector; the first drop line connector connects to the first central trunk; the first drop line connector and the first drop line electrically couple the first solar array to the first or second feeder cable; a second drop line of the plurality of drop lines terminates at a second drop line connector; the second drop line connector connects to the second central trunk; and the second drop line connector and the second drop line electrically couple the second solar array to the first or second feeder cable; wherein a first mold enclosure that encloses at least a portion of a first nexus of the first feeder cable includes: a feeder cable passage that extends completely through the first mold enclosure, the first feeder cable extending through the feeder cable passage along a first longitudinal axis; a first drop line passage that extends at least partially through the first mold enclosure, a first drop line of the plurality of drop lines extending through the first drop line passage along a second longitudinal axis; and a second drop line passage that extends at least partially through the first mold enclosure, a second drop line of the plurality of drop lines extending through the second drop line passage along a third longitudinal axis; and wherein the first and second drop lines are part of a single drop line cable and the first and second drop line passages form a drop line cable passage that extends completely through the first mold enclosure.

8. The solar power system of claim 7, wherein the first feeder cable is configured for electrical connection to an inverter without a combiner box between the first feeder cable and the inverter.

9. The solar power system of claim 7, wherein each mold enclosure completely encloses the corresponding nexus.

10. The solar power system of claim 7, wherein each mold enclosure comprises: an overmold that encloses the corresponding nexus; or an undermold that encloses the corresponding nexus and an overmold that encloses the undermold.

11. The solar power system of claim 7, wherein at least one of the second longitudinal axis or the third longitudinal axis is not parallel to the first longitudinal axis.

12. A lead assembly for use in a solar energy installation, the lead assembly comprising: a first drop line that has a first diameter and that terminates at a drop line connector, the drop line connector configured to connect with a wire harness, the wire harness configured to electrically connect to an array of solar panels; a second drop line that has a second diameter; a feeder cable electrically coupled to the first and second drop lines at a nexus, wherein the feeder cable has a third diameter that is greater than the first diameter and greater than the second diameter; an inline fuse disposed in the first drop line between the nexus and the drop line connector; and a mold structure that at least partially surrounds the nexus, the mold structure defining: first and second openings from which the feeder cable extends from the mold structure in opposite directions, a feeder cable passage extending completely through the mold structure between the first and second openings, the feeder cable extending through the feeder cable passage; a third opening from which the first drop line extends from the mold structure; and a fourth opening from which the second drop line extends from the mold structure; wherein a drop line cable passage extends completely through the mold structure between the third and fourth openings and the first and second drop lines are each part of a single drop line cable that extends completely through the drop line cable passage.

13. The lead assembly of claim 12, wherein the third and fourth openings defined by the mold structure are on opposite sides of the mold structure and the first and second drop lines extend from the third and fourth openings of the mold structure in opposite directions.

14. The lead assembly of claim 12, wherein the feeder cable is configured for electrical connection to an inverter without a combiner box between the feeder cable and the inverter.

15. The lead assembly of claim 12, wherein each mold enclosure completely encloses the corresponding nexus.
 16. The lead assembly of claim 12, wherein the first drop line is configured to be electrically connected to a central trunk of the wire harness, the central trunk having a plurality of branches that are configured to electrically connect to a separate solar panel contained in the array of solar panels.
 17. The lead assembly of claim 12, wherein each mold structure comprises: an overmold that encloses the nexus; or an undermold that encloses the nexus and an overmold that encloses the undermold.
 18. The lead assembly of claim 12, wherein at least one of the second longitudinal axis or the third longitudinal axis is not parallel to the first longitudinal axis.
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